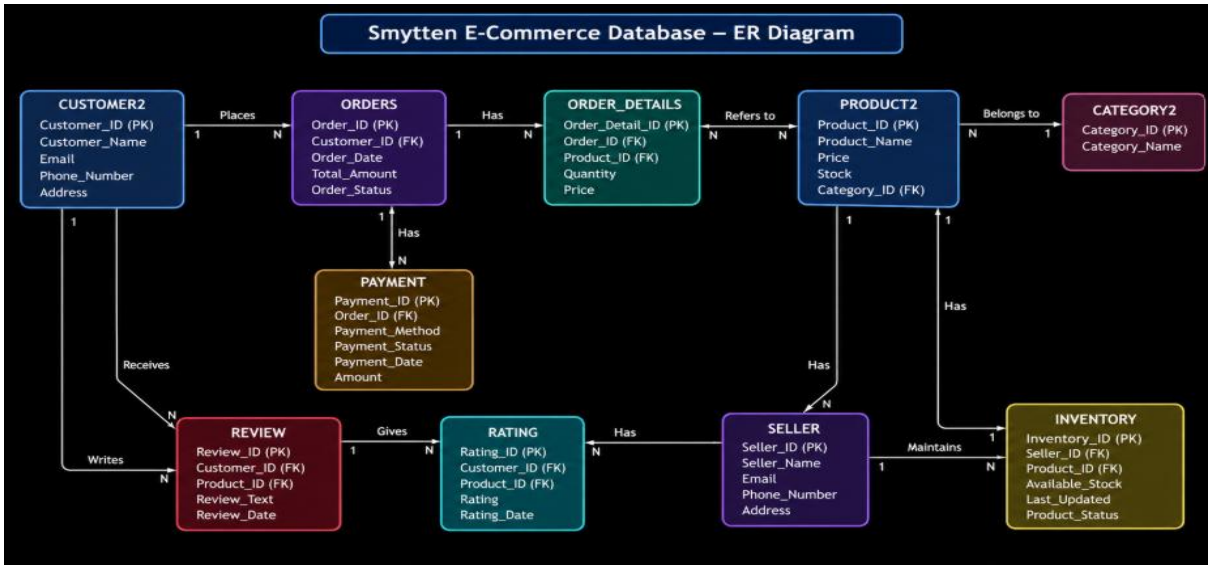
Smytten E-Commerce Database Management System

DATABASE RELATIONSHIP ANALYSIS USING JOINS

CHAPTER-9

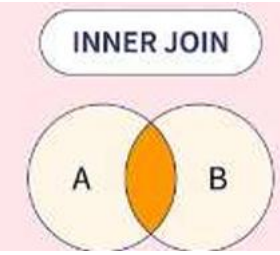
1.Combining Multiple Tables:

The Smytten database contains different tables that store customer, product, order, and payment information. The tables can be combined using JOIN operations to retrieve related information**.**



2. INNER JOIN:

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders:

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  O.TOTAL_AMOUNT,
  6  O.ORDER_STATUS
  7  FROM CUSTOMER2 C
  8  INNER JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  ORDER BY O.ORDER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ ------------------------------
       101
Haripriya
       201 10-SEP-26          899 Delivered

       102
Ananya
       202 11-SEP-26         1299 Shipped
```

2.2 Orders and Payment:

```
SQL> SELECT O.ORDER_ID,
  2  O.CUSTOMER_ID,
  3  O.ORDER_DATE,
  4  O.TOTAL_AMOUNT,
  5  P.PAYMENT_ID,
  6  P.PAYMENT_METHOD,
  7  P.PAYMENT_STATUS,
  8  P.AMOUNT
  9  FROM ORDERS O
 10  INNER JOIN PAYMENT P
 11  ON O.ORDER_ID = P.ORDER_ID
 12  ORDER BY O.ORDER_ID;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID
---------- ----------- --------- ------------ ----------
PAYMENT_METHOD                 PAYMENT_STATUS                    AMOUNT
------------------------------ ------------------------------ ----------
       201          101 10-SEP-26          899         401
UPI                            Success                           899

       202          102 11-SEP-26         1299         402
Credit Card                    Success                          1299

       203          103 11-SEP-26          599         403
Debit Card                     Success                           599
```

2.3 Product and Order_Details:

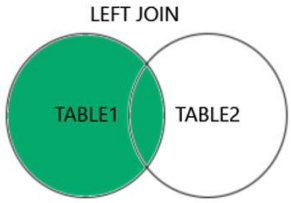
```
SQL> SELECT P.PRODUCT_ID,
  2  P.PRODUCT_NAME,
  3  P.PRICE,
  4  OD.ORDER_ID,
  5  OD.QUANTITY,
  6  OD.PRICE AS UNIT_PRICE
  7  FROM PRODUCT2 P
  8  INNER JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  ORDER BY OD.ORDER_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE   ORDER_ID   QUANTITY UNIT_PRICE
---------- ---------- ---------- ----------
       101
Face Wash
       349        201          2        400

       102
Moisturizer
       399        201          1         99
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders:

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  O.TOTAL_AMOUNT,
  6  O.ORDER_STATUS
  7  FROM CUSTOMER2 C
  8  LEFT JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ ------------------------------
       101
Haripriya
       201 10-SEP-26          899 Delivered

       101
Haripriya
       211 23-SEP-26          899 Shipped
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.

```
SQL> SELECT O.ORDER_ID,
  2  O.ORDER_DATE,
  3  O.TOTAL_AMOUNT,
  4  O.ORDER_STATUS,
  5  C.CUSTOMER_ID,
  6  C.CUSTOMER_NAME,
  7  C.EMAIL
  8  FROM CUSTOMER2 C
  9  RIGHT JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS                    CUSTOMER_ID
---------- --------- ------------ ------------------------------ -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
       201 10-SEP-26          899 Delivered                             101
Haripriya
haripriya@gmail.com

       202 11-SEP-26         1299 Shipped                               102
Ananya
ananya@gmail.com
```

4.1 Display All Orders and Customer Details:

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  P.PRODUCT_ID,
  6  P.PRODUCT_NAME,
  7  OD.QUANTITY,
  8  OD.PRICE AS UNIT_PRICE,
  9  OD.QUANTITY * OD.PRICE AS ITEM_TOTAL,
 10  O.TOTAL_AMOUNT,
 11  O.ORDER_STATUS,
 12  PY.PAYMENT_METHOD,
 13  PY.PAYMENT_STATUS
 14  FROM CUSTOMER2 C
 15  INNER JOIN ORDERS O
 16  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 17  INNER JOIN ORDER_DETAILS OD
 18  ON O.ORDER_ID = OD.ORDER_ID
 19  INNER JOIN PRODUCT2 P
 20  ON OD.PRODUCT_ID = P.PRODUCT_ID
 21  LEFT JOIN PAYMENT PY
 22  ON O.ORDER_ID = PY.ORDER_ID
 23  ORDER BY O.ORDER_ID;
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT PRODUCT_ID
---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS
------------------------------ ------------------------------
       101

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT PRODUCT_ID
---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE ITEM_TOTAL TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ---------- ------------ ------------------------------
PAYMENT_METHOD                 PAYMENT_STATUS
------------------------------ ------------------------------
Haripriya
```

5. Complete Order Details :

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  P.PRODUCT_NAME,
  6  OD.QUANTITY,
  7  OD.PRICE,
  8  O.TOTAL_AMOUNT
  9  FROM CUSTOMER2 C
 10  INNER JOIN ORDERS O
 11  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  INNER JOIN ORDER_DETAILS OD
 13  ON O.ORDER_ID = OD.ORDER_ID
 14  INNER JOIN PRODUCT2 P
 15  ON OD.PRODUCT_ID = P.PRODUCT_ID
 16  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE TOTAL_AMOUNT
---------- ---------- ------------
       101
Haripriya
       211 23-SEP-26
```

6. Customer Purchase History:

This query displays the products purchased by each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  P.PRODUCT_NAME,
  6  OD.QUANTITY,
  7  O.TOTAL_AMOUNT,
  8  PY.PAYMENT_METHOD,
  9  PY.PAYMENT_STATUS,
 10  PY.PAYMENT_DATE
 11  FROM CUSTOMER2 C
 12  INNER JOIN ORDERS O
 13  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 14  INNER JOIN ORDER_DETAILS OD
 15  ON O.ORDER_ID = OD.ORDER_ID
 16  INNER JOIN PRODUCT2 P
 17  ON OD.PRODUCT_ID = P.PRODUCT_ID
 18  LEFT JOIN PAYMENT PY
 19  ON O.ORDER_ID = PY.ORDER_ID
 20  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE DESC;
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_METHOD
---------- ------------ ------------------------------
PAYMENT_STATUS                 PAYMENT_D
------------------------------ ---------
       101

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_METHOD
---------- ------------ ------------------------------
PAYMENT_STATUS                 PAYMENT_D
------------------------------ ---------
Haripriya
```

7. Customer Purchase History with Payment:

```
SQL> SELECT C.CUSTOMER_NAME,
  2  P.PRODUCT_NAME,
  3  SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  4  SUM(OD.QUANTITY * OD.PRICE) AS PRODUCT_SALES_AMOUNT
  5  FROM CUSTOMER2 C
  6  INNER JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  INNER JOIN ORDER_DETAILS OD
  9  ON O.ORDER_ID = OD.ORDER_ID
 10  INNER JOIN PRODUCT2 P
 11  ON OD.PRODUCT_ID = P.PRODUCT_ID
 12  GROUP BY C.CUSTOMER_NAME, P.PRODUCT_NAME
 13  ORDER BY PRODUCT_SALES_AMOUNT DESC;

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY PRODUCT_SALES_AMOUNT
-------------- --------------------
Haripriya
Face Wash
             5                 2000

Kavya
Mascara
             2                 1598
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT PY.PAYMENT_ID,
  2  O.ORDER_ID,
  3  C.CUSTOMER_NAME,
  4  PY.PAYMENT_METHOD,
  5  PY.PAYMENT_DATE,
  6  PY.AMOUNT,
  7  PY.PAYMENT_STATUS
  8  FROM PAYMENT PY
  9  INNER JOIN ORDERS O
 10  ON PY.ORDER_ID = O.ORDER_ID
 11  INNER JOIN CUSTOMER2 C
 12  ON O.CUSTOMER_ID = C.CUSTOMER_ID
 13  ORDER BY PY.PAYMENT_DATE DESC;

PAYMENT_ID   ORDER_ID
---------- ----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_METHOD                 PAYMENT_D     AMOUNT
------------------------------ --------- ----------
PAYMENT_STATUS
------------------------------
       410        210
Aishwarya
Cash on Delivery               19-SEP-26        749
Pending
```

9. Payment Transaction Report:

```
SQL> SELECT
  2       c.Customer_Name,
  3       o.Order_ID,
  4       o.Order_Date,
  5       o.Total_Amount,
  6       py.Payment_Mode,
  7       py.Payment_Amount,
  8       py.Payment_Status
  9  FROM Customer c
 10  INNER JOIN Orders o
 11       ON c.Customer_ID = o.Customer_ID
 12  INNER JOIN Payment py
 13       ON o.Order_ID = py.Order_ID
 14  ORDER BY o.Order_Date DESC;

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE          PAYMENT_AMOUNT
---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
Anitha Kumar
       201 16-SEP-26         2700 UPI                            2500
Successful

Rahul
       202 15-SEP-26         1800 Card                           1800
Successful

CUSTOMER_NAME
```